Exploring the Possibility of Blood Group Prediction from Fingerprint Images Using Deep Learning

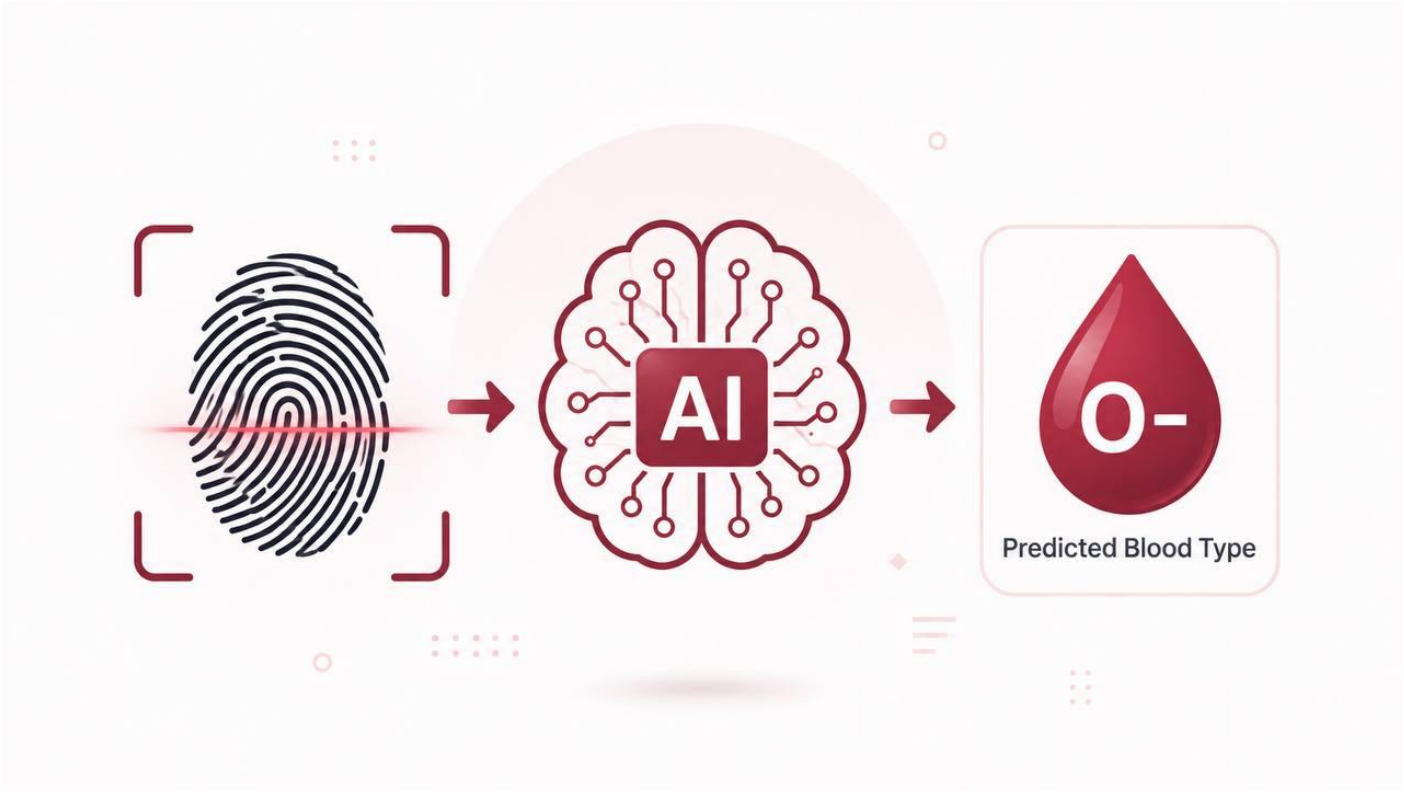
Print2Type

Ghadi Alshahrani | Rana Alshahrani | Wajd Alsubaei
Wateen Almaawi | Wid Alshahrani | Yara Alshahrani

Supervised by: Dr. Ali Faleh Al-Shahrani

Presentation Outline

Introducing Print2Type



Background

Artificial Intelligence

Deep learning enables systems to extract complex patterns from raw data automatically — revolutionizing image analysis across domains.

AI in Medicine

AI is increasingly integrated into medical diagnostics, offering faster, more consistent support for disease detection and clinical decision-making.

Blood Group Identification

A critical medical requirement for transfusions and emergency care, yet current methods depend on invasive lab tests requiring trained personnel.

Problem Statement & Motivation

Problem

- Blood group identification relies on invasive laboratory tests
- Requires trained personnel and specialized equipment
- Limited accessibility in emergency or resource-constrained environments
- Fingerprint data remains largely unexploited in this field
- No intelligent system exists to predict blood groups from fingerprints non-invasively

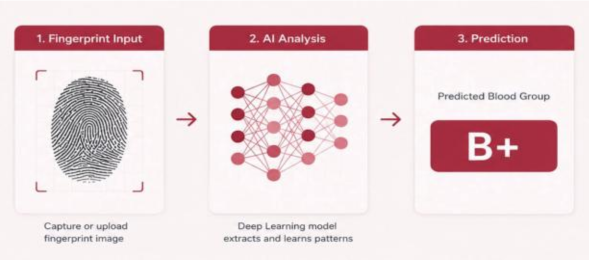
Motivation

- Gap between AI capabilities and medical adoption
- Fingerprints offer a stable, non-invasive biometric input
- Deep learning can detect subtle visual patterns beyond human observation
- Aim to build a complete practical system — not just a theoretical model
- Transform AI research into a deployable, accessible tool

Objectives & Contribution

Project Objectives

1. Explore the feasibility of predicting blood groups from fingerprint images
2. Build and evaluate Print2Type across 8 categories: A+, A−, B+, B−, AB+, AB−, O+, O−
3. Prepare a complete pipeline: preprocessing, training & stratified splitting
4. Assess performance using Accuracy, Precision, Recall, F1-Score, and AUC
5. Deploy Print2Type via Gradio on Hugging Face Spaces



Key Contributions

- Fingerprint images proposed as non-invasive input for blood group prediction
- Print2Type: CNN system for 8-class biometric classification
- End-to-end pipeline: data → training → evaluation → deployment
- Gradio web interface publicly accessible on Hugging Face Spaces
- Highlights challenges and limitations of biometric-based medical AI

Dataset Description

6,000
Total Images

8
Blood Group Classes

Kaggle
Source

224×224
Image Size

Class	Images	%
A+	565	9.4%
A−	1,009	16.8%
AB+	708	11.8%
AB−	761	12.7%
B+	652	10.9%
B−	741	12.3%
O+	852	14.2%
O−	712	11.9%

Sample Fingerprint Image per Blood Group

Data Preprocessing

01

Image Loading

Traverse dataset folders and record all image paths with corresponding blood group labels.

02

Resize & Normalize

Resize to 224×224 px. Normalize pixel values from [0,255] to [0.0,1.0] by dividing by 255.

03

Stratified Splitting

Split: 70% Training (4,200 images) | 30% Test (1,800 images) — preserving class distribution.

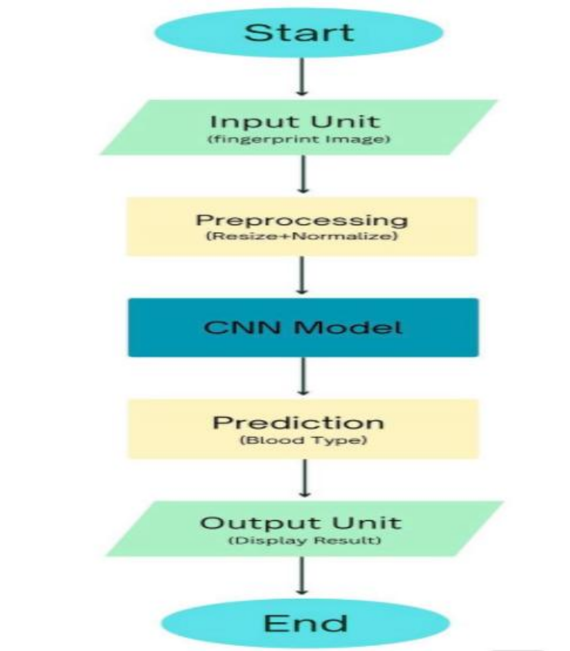
04

Shuffle & Batch

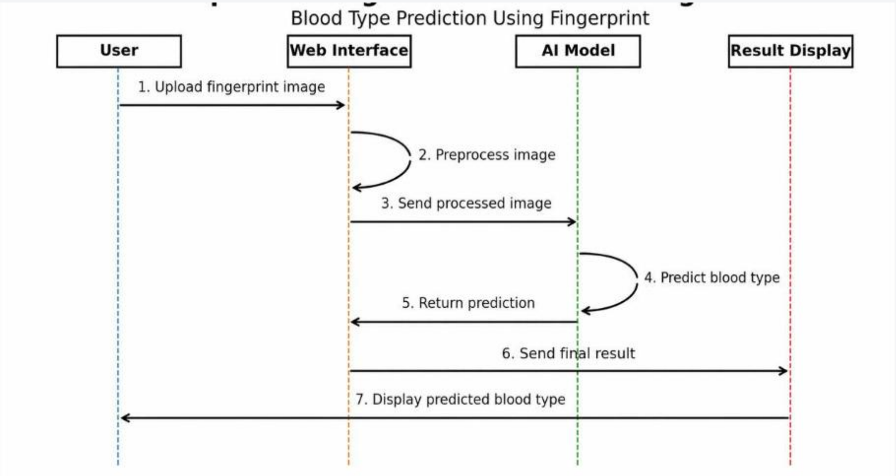
Shuffle training data randomly. Process in batches of 32 for efficient GPU utilization.

Print2Type Model Overview

Print2Type System Flowchart



Sequence Diagram — User Interaction with Print2Type



Print2Type Model Architecture

Layer	Output Shape	Details
Conv2D	(222, 222, 32)	32 filters, 3×3, ReLU
MaxPooling2D	(111, 111, 32)	Pool size 2×2
Conv2D	(109, 109, 64)	64 filters, 3×3, ReLU
MaxPooling2D	(54, 54, 64)	Pool size 2×2
Conv2D	(52, 52, 128)	128 filters, 3×3, ReLU
MaxPooling2D	(26, 26, 128)	Pool size 2×2
Flatten	(86,528)	—
Dense	(128)	ReLU activation
Dropout	(128)	Rate = 0.5
Dense (Output)	(8)	Softmax — 8 blood group classes

Architecture
Custom CNN (from scratch)

Optimizer
Adam

Loss Function
Sparse Categorical Cross-Entropy

Total Params
11,189,992 (42.61 MB)

Output
Softmax — 8 classes

Evaluation Strategy

Accuracy

`(TP+TN) / (TP+TN+FP+FN)`

Overall proportion of correctly classified samples

Precision

`TP / (TP + FP)`

Correctly predicted positives among all predicted positives

Recall

`TP / (TP + FN)`

Actual positives correctly identified by the model

F1-Score

`2 × (P × R) / (P + R)`

Harmonic mean of precision and recall

AUC-ROC

`One-vs-Rest approach`

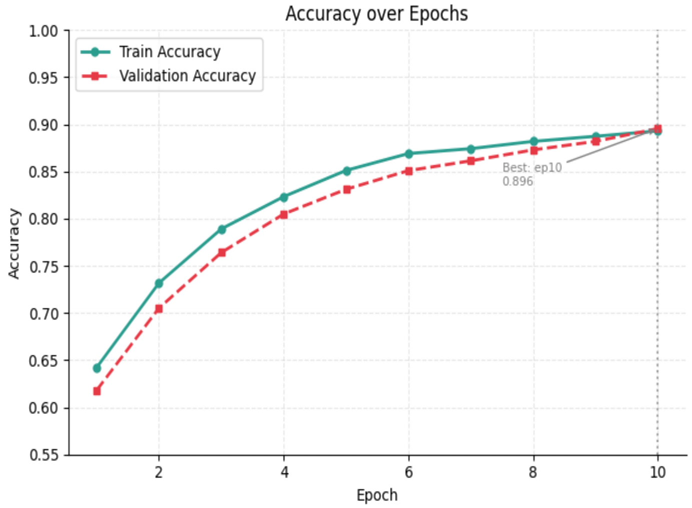
Discriminative ability across all classification thresholds

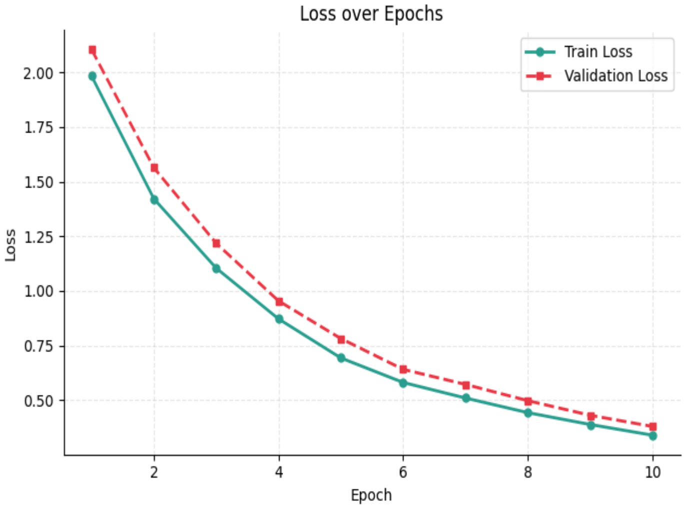
Training Configuration & Results

Epochs	10
Batch Size	32
Optimizer	Adam
Training Set/Test Set	4,200 images / 1,800 images

Training accuracy improved steadily from 64% (epoch 1) to 89% (epoch 10).

Loss decreased consistently with no significant overfitting observed.





Classification Report

Class	Precision	Recall	F1-Score	Support
A+	0.87	0.88	0.87	169
A−	0.91	0.89	0.90	303
AB+	0.90	0.89	0.89	212
AB−	0.91	0.89	0.90	228
B+	0.88	0.89	0.88	196
B−	0.89	0.89	0.89	222
O+	0.89	0.91	0.90	256
O−	0.87	0.88	0.88	214
Macro Avg	**0.89**	**0.90**	**0.89**	**1,800**

Overall Accuracy

89.56%

Best F1-Score

0.90 (A−, AB−, O+)

F1-Score Range

0.87 – 0.90

No class bias

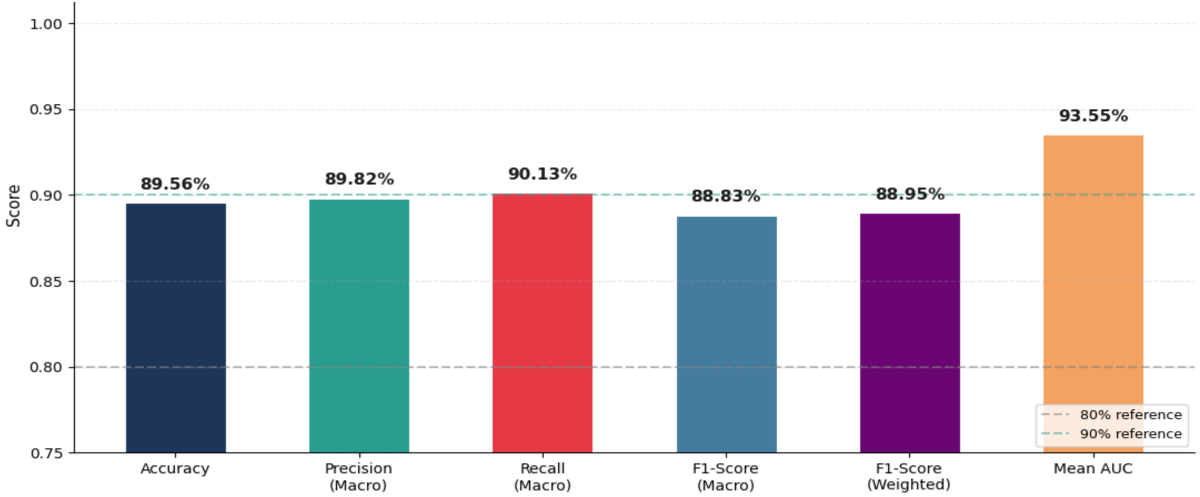
Balanced across all 8

Overall Performance Metrics

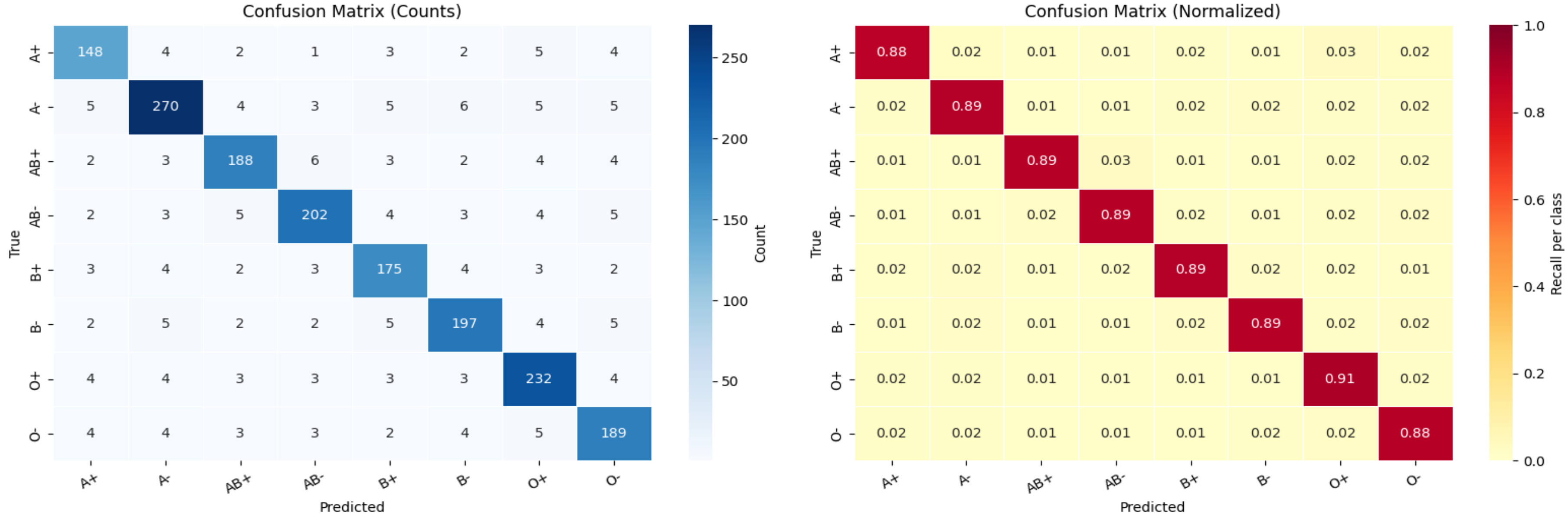
Accuracy	Precision (Macro)	Recall (Macro)
89.56%	89.82%	90.13%

F1-Score (Macro)	F1-Score (Weighted)	Mean AUC (OvR)
88.83%	88.95%	0.9355

Overall Model Performance Summary



Confusion Matrix Analysis

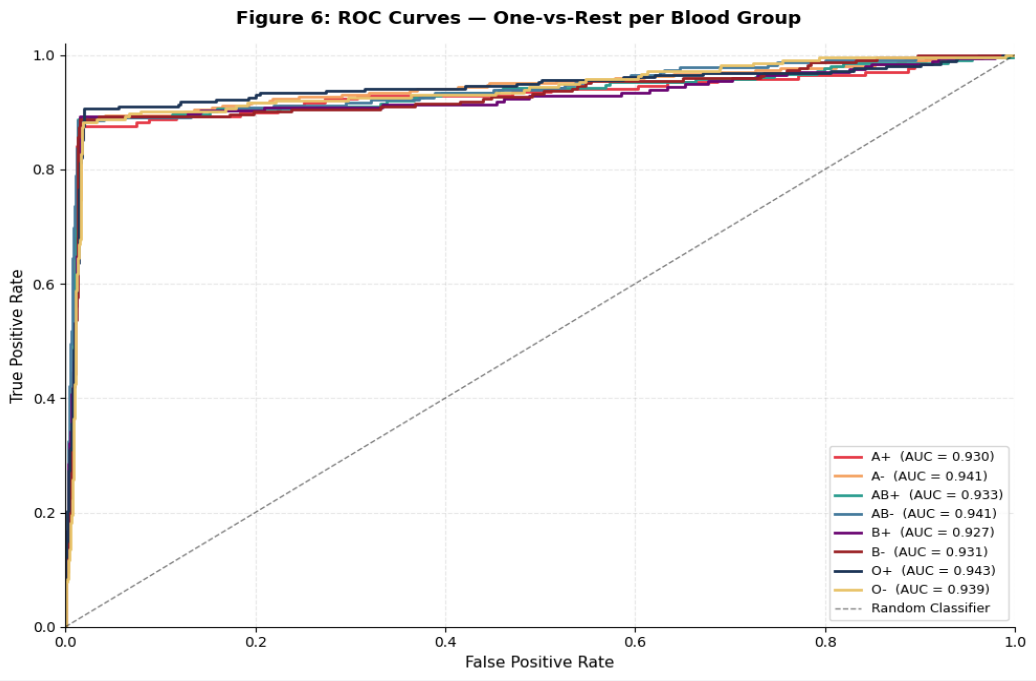


✓ **Strong diagonal values across all 8 classes**

✓ **Recall per class: 0.88 – 0.91 (no dominant error pattern)**

✓ **Misclassifications evenly distributed — no systematic confusion**

ROC Curves — One-vs-Rest



Class	AUC
A+	0.930
A−	0.941
AB+	0.933
AB−	0.941
B+	0.927
B−	0.931
O+	0.943
O−	0.939
Mean AUC	**0.9355**

All classes achieved AUC > 0.92, confirming strong discriminative ability across all decision thresholds.

Web Application — Print2Type

Framework
Gradio — interactive AI web apps

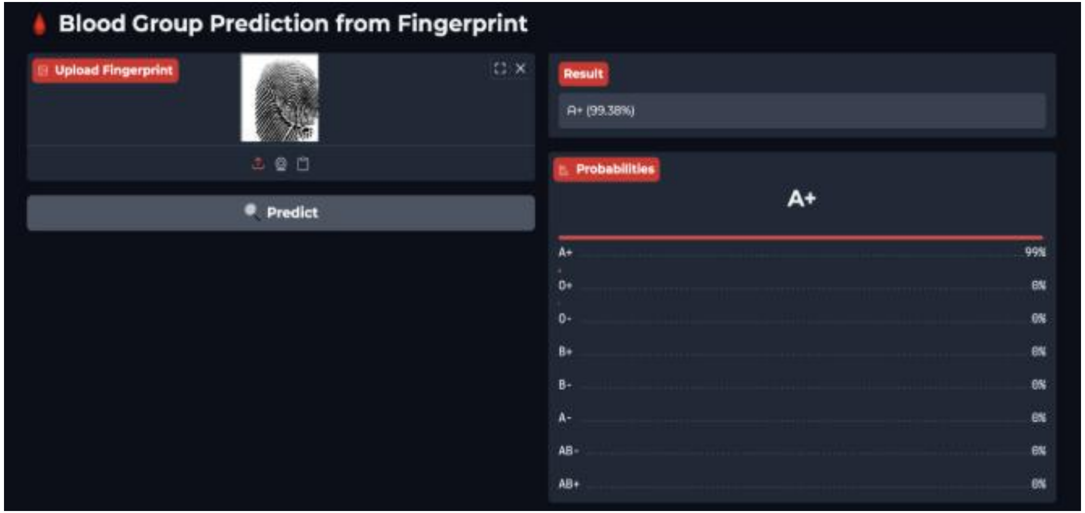
Input
User uploads fingerprint image

Output
Predicted blood group + confidence scores

Hosting
Hugging Face Spaces

Access
Public URL — no installation required

Sample Result
A+ predicted with 99.38% confidence



Comparison: Print2Type vs. Vision Transformer (ViT)

Print2Type (CNN)

89.56%

Accuracy

Balanced, stable training

No overfitting observed

Strong generalization on 6,000 images

Vision Transformer (ViT)

78.08%

Accuracy

Clear overfitting observed

Train accuracy ~95%

Val accuracy stabilized at ~78%

Conclusion: For a dataset of 6,000 images, CNN-based Print2Type generalizes better than ViT, which typically requires significantly larger datasets.

Limitations & Challenges

Limited Dataset

Only 6,000 images — may not capture full diversity of fingerprint patterns.

Computational Constraints

Trained on Google Colab — limited epochs, batch size, and model complexity.

Image Quality Dependency

Blurry or poorly captured images reduce prediction accuracy.

Label Display Issue

Early version omitted +/− signs. Resolved by correcting the label mapping.

Deployment Limitation

Gradio temporary links expire. Resolved by hosting on Hugging Face Spaces.

Print2Type is an experimental prototype — results should not be considered a substitute for professional medical diagnosis.

Future Work

Larger Dataset

Train on a larger, more diverse dataset to improve generalization across fingerprint pattern types.

Advanced Architectures

Explore transfer learning with EfficientNet or ResNet for improved classification accuracy.

Image Quality Enhancement

Apply contrast enhancement and noise reduction to improve robustness to low-quality inputs.

Image Quality Warning

Add automatic detection to notify users when uploaded image quality is insufficient for reliable prediction.

Explainability (Grad-CAM)

Visualize regions influencing predictions to increase transparency and trust in medical contexts.

Health System Integration

Connect Print2Type with healthcare platforms for broader practical applicability in clinical environments.

Conclusion

Q

Can CNN predict blood groups from fingerprints?

→ **Yes — Print2Type achieved 89.56% accuracy and 0.9355 mean AUC.**

Q

Is performance balanced across all 8 classes?

→ **Yes — F1-scores range 0.87–0.90 with no class bias.**

Q

Is the web interface practical?

→ **Yes — real-time prediction with 99.38% confidence demonstrated.**

Print2Type demonstrates the feasibility of using fingerprint images as a non-invasive biometric input for blood group prediction through deep learning.

Print2Type Demo

Thank You
For Listening !

We welcome your questions

Print2Type Team

University of Bisha | 2026